

RELIABILITY-LAB TECHNICAL REPORT

Stochastic Structural Reliability & Provenance Certification

Metric	Value	Standard Compliance
Failure Probability (Pf)	1.1300e-03	ISO 2394 / DNV-RP-C210
Reliability Index (Beta)	3.054	Class Target Beta >= 3.8
Coefficient of Variation (CoV)	16.08%	Moderate
Simulation Method	Subset Simulation	Rare-Event Estimator
Total Function Evaluations	3000	High-Fidelity Verified

1. Limit-State & Model Specification

Performance Function: $g(X) = R - S$
Joint Distribution Copula: Gaussian Copula with Marginal Distributions

2. Subset Simulation Level Diagnostics

Level	Threshold (b_k)	Conditional P(F_k)	Evaluations	Acceptance Rate
0	42.1522	0.1000	1000	100.0%
1	19.2559	0.1000	1000	70.5%
2	0.0000	0.1130	1000	70.7%

3. Provenance & Reproducibility Certification

This technical document was automatically synthesized by Reliability-Lab v0.1.0 with strict seed reproducibility guarantees and numerical verification.